# Artificial Intelligence Basics

A concise introduction to AI, machine learning, deep learning, and generative AI.

## 1. What is Artificial Intelligence?

Artificial Intelligence (AI) is a field of computer science focused on building systems that perform tasks commonly associated with human intelligence.

Examples include understanding language, recognizing images, making recommendations, and planning actions.

## 2. Machine Learning

Machine Learning (ML) is a branch of AI in which a model learns patterns from data to make predictions or decisions.

Classification predicts a category, such as spam or not spam. Regression predicts a numerical value, such as a house price.

## 3. Supervised Learning

Supervised learning uses labeled examples. Each training example contains input data and a known target or correct output.

The model learns a relationship between inputs and targets, then predicts targets for new inputs.

## 4. Unsupervised Learning

Unsupervised learning finds patterns in data without provided target labels.

Clustering groups similar data points. K-means is a commonly used clustering algorithm.

## 5. Deep Learning

Deep Learning is a part of machine learning that uses neural networks with multiple layers.

It is useful for complex patterns in images, audio, text, and other large datasets.

## 6. Generative AI

Generative AI creates new content such as text, images, audio, video, or code based on patterns learned from data.

Large language models are generative AI systems that can process and generate text.